# Jingming Liang (Brighton)

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## EDUCATION

**Nankai University, School of Computer Science** Tianjin, China
B.Eng. in Computer Science and Technology (expected) Sep. 2023 - Jun. 2027

- Minors: Finance and Criminal Law. University of Macau exchange programme admission.
- **Research interests:** AI infrastructure, LLM inference systems, heterogeneous computing, and on-device AI.
- **Selected coursework:** Computer Architecture, Parallel Programming, Operating Systems, Computer Networks, Machine Learning, Deep Learning, and Computer Vision.

## RESEARCH EXPERIENCE

**Diagnosing and Mitigating Noun-Driven Routing Bias in Multimodal MoE**
Research project | Routing probe design and experimental evaluation Apr. - Jun. 2026

- Developed **CMRD**, a training-free routing probe, to diagnose noun-driven visual routing shifts in MoE-LLaVA-Qwen; localized a shared **layer-10 routing hotspot** using same-question controls.
- Evaluated Noun-Blind Routing and Two-Stage Verification on **POPE-9000**. Two-Stage Verification reduced false positives by **59.5%** and improved accuracy from **81.7% to 88.2%**, with increased false negatives.

**Long-Context Reasoning and KV-Cache Optimization**
Research project | Query geometry analysis and inference-time aggregation May - Jul. 2026

- Identified stable **pre-RoPE anisotropy** across **1,152 query heads in Qwen3-8B**. Developed **TailTri**, replacing mean aggregation with Log-Sum-Exp to improve long-context reasoning.
- Improved the project's **AIME 2025** evaluation result from **6/30 to 11/30** in the project experiments.

**K-LoRA++: Adaptive Weight Scheduling for Image Diffusion Models**
Competition research project | National Third Prize, Challenge Cup Apr. - Oct. 2025

- Extended **K-LoRA (Ouyang et al., CVPR 2025)** for training-free subject/style LoRA fusion; formulated and evaluated **8 stage-aware scaling schedules** across diffusion timesteps.
- Compared schedules using derivative analysis and cross-domain experiments. A linear schedule gave the best content/style balance among the evaluated schedules; contributed to the interactive demo.

## SELECTED SYSTEMS PROJECTS

**GPU Optimization for All-Pairs Shortest Path**
Individual competitor | CCF-TCARCH National Champion Aug. - Sep. 2025

- Designed a **three-stage blocked Floyd-Warshall pipeline** with shared-memory reuse, kernel fusion, dual streams, and double buffering; reduced host-to-device transfer complexity from **$O(V^2)$ to $O(E)$**.
- Reduced end-to-end runtime from **11.50 s to 5.42 s** on official hidden datasets: **52.9% lower runtime (2.12× speedup)**. Won the national championship as an individual competitor.

**On-Device AIOS on RK3588 and StarryOS**
Team captain | National Third Prize, Operating System Design Contest May - Aug. 2026

- Integrated **UVC acquisition, RKNN/NPU inference, object detection, and robotic-arm control** for a tennis-ball retrieval robot on RK3588 and StarryOS; diagnosed RKNPU and SMP scheduling bottlenecks.
- Reduced **rknn_run latency from 817 to 211 ms** and improved four-thread speedup from **0.96× to 3.45×** by fixing logging stalls and single-core task placement. Contributed **30+ merged pull requests to StarryOS/tgoskits**.

**LLM Inference Serving on DCU Accelerators**
Individual competitor | National Third Prize, Xiandao Cup Jul. - Aug. 2026

- Profiled **Qwen3.5-27B + vLLM** on DCU accelerators for long-context inference; analyzed prefill/decode bottlenecks in full attention, **Gated DeltaNet (GDN)**, and paged key-value (KV) cache operations.
- Optimized kernels and memory access with **FlashAttention, Triton GEMV, and CUDA Graph**; adapted **TP=2** and Custom AllReduce for two-device serving, placing **12th nationally**.

**MoE Language Model Inference Optimization**
National Third Prize, Xiandao Cup May - Nov. 2025

- Profiled **Mixtral-8×7B on DCU accelerators** to identify expert-scheduling, attention, and memory bottlenecks; optimized multi-accelerator inference through dual batching, pipelining, and dynamic load balancing.
- Reduced average latency from **6.6 to 2.71 s/sample** in the project measurements, placing **10th nationally**.

## INTERNSHIP EXPERIENCE

**UniGrow, Technical Development** Jun. - Oct. 2026
AI Agent Full-Stack Engineer Intern
- Contributed to development and production delivery across **AI agents, RAG, mobile apps, backend services, and CI/CD**; handled deployment and Android/iOS releases. The product reached **1,000+ registered users within two weeks** of launch.

**Hejun Consulting, Environmental Services** Jul. - Sep. 2025
Business Analysis Intern | Tianjin, China
- Cleaned 2016-2024 financial data and regional water-utility market data; built market-size estimates, competitor benchmarks, and regional-attractiveness indicators. Delivered a **30,000+ word industry analysis report**.

**ChinaSoft International, LLM Team** Jun. - Jul. 2025
LLM Development Intern | Tianjin, China
- Built an e-commerce **RAG** pipeline with data extraction, knowledge-graph/vector-store integration, and retrieval tuning; fine-tuned a **Qwen digital persona with 4-bit QLoRA on an RTX 4060 (8 GB)** and deployed it on QQ.

**Guotai Junan Securities, Wealth Management** Jun. - Jul. 2024
Quantitative Research Assistant | Remote
- Analyzed ten years of data across ten asset classes; applied **Markowitz mean-variance optimization** to estimate the efficient frontier and maximum-Sharpe portfolio, and examined corner-solution overfitting and dynamic adjustments.

## ADDITIONAL PROJECTS AND OPEN SOURCE

- **SparseTSF-FFT (Sep. - Dec. 2024):** Migrated SparseTSF from PyTorch to **MindSpore** and replaced local convolutions with learnable FFT filtering; improved long-horizon forecasting in the tested ETTh1/ETTh2 settings.
- **RaceRadar (Nov. 2025 - present):** Built a multi-source competition crawler with scheduled GitHub Actions and a native **Swift/Xcode iOS app** for discovery, prioritization, and sharing. App Store release in preparation.
- **Lamplighter Commons:** Founded an open-source organization; created **Triangle Light Maze Solver**, an interactive dynamic-programming solver for online assessment puzzles (**53 stars, 7 forks**, Oct. 18, 2026). Contributor to **StarryOS / tgoskits**.

## SELECTED HONORS AND AWARDS

- **National Champion**, 8th CCF-TCARCH Computer Architecture Challenge, individual competitor (2025).
- **National Third Prize**, Operating System Design Contest, OS Function Challenge Track (team captain, 2026); Intelligent Computing Innovation Design Contest / Xiandao Cup (2025, 2026).
- **Third Prize and Special Contribution Award**, CANN-HCCL Communication Library Innovation Competition (2026).
- **National Third Prize**, 19th Challenge Cup National Undergraduate Extracurricular Academic Science and Technology Works Competition (2025); **Grand Prize**, Challenge Cup Tianjin competition (2025).
- **Provincial First Prize** (2025) and **Provincial Second Prize** (2024), China Undergraduate Mathematical Contest in Modeling (CUMCM); **Honorable Mention**, Mathematical Contest in Modeling (MCM, 2025), all as team captain.
- **National First Prize**, 3rd National College Student Data Analysis Popular Science Competition (2024).
- **Second Place**, TRAE on Campus Vibe Coding Workshop (2025); **Special Prize**, Pioneer Cup Intelligent Computing Design Competition (2025).

## SCHOLARSHIPS

- **Gongneng Scholarship**, Nankai University, **CNY 5,000** (Oct. 2026).
- **Academic Progress** (2025), **Social Welfare** (2024), and **Student Service** (2024, 2025) Scholarships, Nankai University.

## TEACHING AND EXTRACURRICULAR EXPERIENCE

- **Teaching Assistant, Computer Hardware Fundamentals** (Sep. 2025 - Jan. 2026): supported lectures, assignment evaluation, and student Q&A; recognized as an Outstanding Teaching Assistant.
- **Class Monitor and Youth League Branch Deputy Secretary** (Sep. 2023 - present): coordinated class activities and peer support; Outstanding Student Cadre (2024, 2025). **530 hours** of peer learning support, **249.9 hours** of volunteer service, and **153 hours** of social practice.
- **CS and AI Educational Outreach** (Aug. 2024 - present): created study notes and learning resources on Rednote; **6,211 followers, 3M+ views, and 156k likes and saves** (Oct. 2026).

## SKILLS AND INTERESTS

- **Technical:** C++, Python, SQL, MATLAB, Swift/Xcode, PyTorch, MindSpore, vLLM, Triton, FastAPI, LangChain, Git/GitHub Actions, and LaTeX.
- **Languages:** Chinese (native); English: College English Test **Band 4 (CET-4): 583; Band 6 (CET-6): 521**.
- **Interests:** Violin and classical guitar (**Grade 10 in both, China Conservatory of Music**), volleyball, marathon running, and independent travel.